

Hypersingular Riesz-energy constants on surfaces from triangular-lattice universal optimality

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October 6, 2026

Abstract

We deduce the exact two-dimensional hypersingular minimal Riesz-energy constant for every real $s > 2$ from OpenAI's recently released triangular-lattice universal-optimality theorem. The constant is the Epstein sum of the covolume-one triangular lattice. An explicit gap-periodization argument relates the finite square problem to the source theorem's centered-disk lower energy, with an inter-box error of order $N^{1-s/2}$ for each fixed gap parameter. A lattice crop gives the matching upper bound. The corrected Hardin–Saff universality theorem then gives the ordered-pair leading asymptotic on every smooth compact embedded surface of positive area, including surfaces with smooth boundary, for ambient Euclidean distance. This resolves the two-dimensional $s > 2$ case of the explicit lattice-constant conjecture recorded by Brauchart, Hardin and Saff. The universal-optimality breakthrough, finite-to-infinite reduction and manifold theorem are inherited results; this note gives a self-contained account of the known passage and records its planar surface consequence.

1 Statement, provenance and conventions

Put $b = \sqrt{3}/2$ and

$$\Lambda = b^{-1/2}\{j(1, 0) + k(1/2, b) : j, k \in \mathbb{Z}\}, \quad Z_s = \zeta_\Lambda(s) = \sum_{v \in \Lambda \setminus \{0\}} |v|^{-s}.$$

The lattice has covolume one. For a compact set $K \subset \mathbb{R}^p$, define

$$\mathcal{E}_s(K, N) = \min_{x_1, \dots, x_N \in K} \sum_{i \neq j} |x_i - x_j|^{-s},$$

with collisions assigned energy $+\infty$. All pair sums in this note are *ordered*. Distances are Euclidean distances in the ambient space, not intrinsic geodesic distances. The Hausdorff measure $\mathcal{H}^2$ is normalized to agree with planar Lebesgue area.

Theorem 1. *For every real $s > 2$,*

$$C_{s,2} := \lim_{N \rightarrow \infty} \frac{\mathcal{E}_s([0, 1]^2, N)}{N^{1+s/2}} = Z_s. \quad (1)$$

If $M \subset \mathbb{R}^p$ is a smooth compact embedded two-dimensional manifold, with or without smooth boundary, and $A = \mathcal{H}^2(M) > 0$, then

$$\mathcal{E}_s(M, N) \sim Z_s A^{-s/2} N^{1+s/2}. \quad (2)$$

In particular, the unit sphere in $\mathbb{R}^3$, with chordal distance, has coefficient $Z_s/(4\pi)^{s/2}$.

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The candidate constant in dimension two goes back to Kuijlaars and Saff [1, Conjecture 1], whose unordered-pair coefficient must be doubled in our convention; Brauchart–Hardin–Saff [3, Conjecture 2] state the formula $C_{s,d} = \text{covol}(\Lambda_d)^{s/d} \zeta_{\Lambda_d}(s)$ for $s > d$ in dimensions 2, 4, 8, 24. We establish only its $d = 2$, $s > 2$ part. The result concerns the leading minimum value, not microscopic crystallization or uniqueness of finite minimizers. It makes no new assertion for $s \leq 2$ or about a second-order term.

The new pivotal input is OpenAI’s infinite-plane theorem [7, Theorem 1.1], also proved in its atomic companion [8, Theorem 1.1]. We use the exact source version at repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. For a locally finite $\mathcal{C} \subset \mathbb{R}^2$, let B_R be the closed disk centered at zero and $n_{\mathcal{C}}(R) = \#(\mathcal{C} \cap B_R)$. The source uses centered density $n_{\mathcal{C}}(R)/(\pi R^2) \rightarrow 1$ and

$$\mathcal{H}_s(\mathcal{C}) = \liminf_{R \rightarrow \infty} \frac{1}{n_{\mathcal{C}}(R)} \sum_{\substack{x, y \in \mathcal{C} \cap B_R \\ x \neq y}} |x - y|^{-s}. \quad (3)$$

The imported theorem says that $\mathcal{H}_s(\mathcal{C}) \geq Z_s$ for every such configuration. Indeed $g(t) = t^{-s/2}$ satisfies its smooth nonnegative complete-monotonicity hypotheses, since

$$(-1)^r g^{(r)}(t) = (s/2)_r t^{-s/2-r} > 0 \quad (t > 0, r \geq 0).$$

Here $(s/2)_0 = 1$. Section 2 proves the exact finite transfer; only the square-periodic special case of the imported theorem is needed. The finite/periodic asymptotic equivalence was proved by Hardin, Saff and Simanek [5, Theorem 3.2]. Hardin, Leblé, Saff and Serfaty [6, Proposition 3.1] identify the thermodynamic minimum with $C_{s,d}$; their published note explicitly draws the analogous $d = 8, 24$ consequence of universal optimality. Periodic centered-ball limits also appear in Cohn–Kumar [4, Lemma 9.1]. The passage proved below is this established reduction in the present conventions. This is an explicit newly available consequence of the planar theorem, not an independently discovered base breakthrough. We do not claim an independent proof of planar universal optimality, invention of these reductions, or first public priority for the corollary.

2 Periodic disk limits and a gap-periodization bound

We first justify the disk-energy limit for the periodic configurations used below, so there is no implicit change of averaging convention.

Lemma 2. *Let $L > 0$, $q \geq 1$, and $F = \{f_1, \dots, f_q\} \subset [0, L)^2$ consist of distinct points. For $\mathcal{C} = F + L\mathbb{Z}^2$ and $s > 2$, the centered density is q/L^2 , and the lower limit in (3) is an ordinary finite limit equal to*

$$\frac{1}{q} \sum_{i=1}^q h_i, \quad h_i = \sum_{\substack{1 \leq j \leq q, k \in \mathbb{Z}^2 \\ (j,k) \neq (i,0)}} |f_i - f_j - Lk|^{-s}. \quad (4)$$

Proof. Representations are unique in the half-open fundamental square, and the set is locally finite. For $m = \|k\|_{\infty} \geq 2$, each denominator in the i th row is at least $L(m-1) \geq Lm/2$. There are $8m$ such translations, so the tail is bounded by $8q(2/L)^s \sum_{m \geq 2} m^{1-s} < \infty$. The remaining finitely many denominators are nonzero. Thus $h_i < \infty$; all rearrangements of these nonnegative sums are justified by Tonelli.

Let $n_i(R) = \#((f_i + L\mathbb{Z}^2) \cap B_R)$ and $d = L\sqrt{2}/2$. The side- L cells centered at these points give

$$\pi(R-d)^2/L^2 \leq n_i(R) \leq \pi(R+d)^2/L^2 \quad (R > d).$$

Their union contains B_{R-d} and is contained in B_{R+d} . Consequently $n_i(R) = \pi R^2/L^2 + O(R/L + 1)$ and $n_i(R)/n_C(R) \rightarrow 1/q$.

Write S_R for the numerator in (3) and let $h_i^{(T)}$ be the row restricted to displacements of length at most T . For $R > T$, completing outgoing rows and, respectively, retaining only rows from B_{R-T} give

$$\sum_i n_i(R-T)h_i^{(T)} \leq S_R \leq \sum_i n_i(R)h_i.$$

Divide by $n_C(R)$ and take $R \rightarrow \infty$ at fixed T . Then $T \rightarrow \infty$ gives $h_i^{(T)} \uparrow h_i$ by monotone convergence, proving (4). This also proves the asserted density. $\square$

For $\lambda > 0$, direct substitution of R/λ in the definitions gives

$$\text{dens}(\lambda\mathcal{C}) = \lambda^{-2} \text{dens}(\mathcal{C}), \quad \mathcal{H}_s(\lambda\mathcal{C}) = \lambda^{-s} \mathcal{H}_s(\mathcal{C}). \quad (5)$$

Let $a_N = N^{-1-s/2} \mathcal{E}_s([0, 1]^2, N)$, and distinguish the Riemann zeta function $\zeta_R(u) = \sum_{m \geq 1} m^{-u}$ from the lattice sum.

Proposition 3. *For each $s > 2$, $N \geq 1$ and $\varepsilon > 0$,*

$$a_N \geq (1 + \varepsilon)^{-s} Z_s - 8\varepsilon^{-s} \zeta_R(s-1) N^{1-s/2}. \quad (6)$$

Proof. Take any distinct N -point set $X = \{x_i\} \subset [0, 1]^2$, set $y_i = \sqrt{N}x_i$, $L = (1 + \varepsilon)\sqrt{N}$, and repeat $F = \{y_i\}$ by $L\mathbb{Z}^2$. The strict gap makes $F \subset [0, L)^2$, even when X contains boundary points, and the density is $(1 + \varepsilon)^{-2}$. By Lemma 2, its energy per point is

$$N^{-1-s/2} \sum_{i \neq j} |x_i - x_j|^{-s} + I_{N,\varepsilon}, \quad I_{N,\varepsilon} = \frac{1}{N} \sum_{i,j=1}^N \sum_{k \neq 0} |y_i - y_j - Lk|^{-s}.$$

The cross-cell sum includes $i = j$ for nonzero translations. For $m = \|k\|_\infty \geq 1$, a coordinate with $|k_a| = m$ gives

$$|y_i - y_j - Lk| \geq \sqrt{N}((1 + \varepsilon)m - 1) \geq \varepsilon\sqrt{N}m.$$

Counting the $8m$ translations in that shell yields the uniform bound

$$0 \leq I_{N,\varepsilon} \leq 8\varepsilon^{-s} \zeta_R(s-1) N^{1-s/2}. \quad (7)$$

No separation or occupancy bound within X was used.

Compress the repeated configuration by $\lambda = (1 + \varepsilon)^{-1}$. It now has centered density one. Equations (4)–(5) and the imported universal-optimality theorem give

$$Z_s \leq (1 + \varepsilon)^s \left[N^{-1-s/2} \sum_{i \neq j} |x_i - x_j|^{-s} + I_{N,\varepsilon} \right].$$

Combine with (7) and minimize over X to prove (6). Finite minimizers exist by lower semicontinuity on the compact product after assigning $+\infty$ to collisions. $\square$

For fixed $\varepsilon > 0$, the error in (6) tends to zero because $s > 2$. Hence

$$\liminf_{N \rightarrow \infty} a_N \geq (1 + \varepsilon)^{-s} Z_s, \quad \liminf_{N \rightarrow \infty} a_N \geq Z_s$$

after $\varepsilon \downarrow 0$. The limit order is the disk radius first at each fixed N, ε , then $N \rightarrow \infty$ at fixed ε , then $\varepsilon \downarrow 0$. There is no uniform-in- N disk-limit assumption and no periodization with zero gap.

3 The matching lattice upper bound

The sum Z_s converges for $s > 2$: if $\Lambda = B\mathbb{Z}^2$ with B invertible, then $|Bk| \geq c\|k\|_\infty$ for some $c > 0$, and $Z_s \leq 8c^{-s} \sum_{m \geq 1} m^{1-s} < \infty$. Take a bounded fundamental parallelogram P of Λ , contained in $[-D, D]^2$ for some $D > 0$. Its translates tile the plane up to cell boundaries and each has area one. For $t \geq 2D$, every point of $[D, t - D]^2$ lies in a cell whose lattice representative belongs to $[0, t]^2$. Therefore

$$\#(\Lambda \cap [0, t]^2) \geq (t - 2D)^2.$$

With $t_N = \sqrt{N} + 2D$, select exactly N points Y_N from this crop. Each outgoing row of the finite set is at most the full lattice sum Z_s . Scale by $1/t_N$ into the unit square to obtain

$$\mathcal{E}_s([0, 1]^2, N) \leq t_N^s N Z_s, \quad a_N \leq Z_s \left(1 + \frac{2D}{\sqrt{N}}\right)^s.$$

Thus $\limsup a_N \leq Z_s$, proving (1). Selecting an exact- N subset avoids any assumption that a lattice crop can be chosen with exactly N sites. No tail is discarded with an uncontrolled error.

4 Transfer to surfaces and verification scope

Hardin and Saff [2, Theorem 2.4] prove that, for $s > d$, a compact d -rectifiable manifold $K \subset \mathbb{R}^{d'}$ of positive $\mathcal{H}^d$ measure satisfies

$$\lim_{N \rightarrow \infty} \frac{\mathcal{E}_s(K, N)}{N^{1+s/d}} = \frac{C_{s,d}}{\mathcal{H}^d(K)^{s/d}}.$$

Their definition (19) is a finite union $K = \bigcup_k \varphi_k(K_k)$, where each $K_k \subset \mathbb{R}^d$ is compact and φ_k is bi-Lipschitz on an open set $G_k \supset K_k$. The theorem uses ambient Euclidean distance, ordered pairs, and the unit-cube-normalized constant. We use the corrected proof, with the addendum included in the cited arXiv version, rather than the uncorrected neighborhood assertion in Proposition 8.3.

A smooth compact embedded surface with smooth boundary has finitely many such compact chart pieces. At a boundary point, its smooth chart on a half-disk extends to an open planar neighborhood; after shrinking, the extension is bi-Lipschitz by the full-rank derivative and local graph representation. The pieces on the half-disk remain compact subsets of that open neighborhood. Thus the $s > d$ theorem allows boundary here. The extra containment in a C^1 manifold without boundary in the source's $s = d$ clause is not needed for $s > 2$.

For completeness, the $s > d$ addendum's neighborhood lemma is used only inside a *compact* regular thickening: cover each chart piece by finitely many closed cubes contained in G_k and take the union of their bi-Lipschitz images. This is a compact d -regular set containing K . Consequently the closure of each relative neighborhood is compact and its positive-distance step is valid. No unrestricted unbounded-neighborhood version of that lemma is invoked. Applying the theorem with $d = 2$ and (1) proves (2); the sphere has area 4π .

The accompanying dependency ledger and audits record an independent analytic review of the needed parts of both OpenAI proofs, source hashes, and reproduced finite certificates. The main manuscript's two interval checkers completed all 37,310 Bernstein and ten tail comparisons and the scalar assertions; the atomic finite checker also passed. These checks validate explicit inequalities only; the analytic reductions, infinite corrections, global signs and density/mixture transfer were separately reviewed. The actual family-090 Lean source statements were inspected for semantic agreement. A complete build of the pinned compatible library was not reproduced, and this note is not claimed to be formally verified.

Disclosure. AI tools were used extensively in research, drafting, computation and verification. The cited upstream manuscripts are supplied as OpenAI-authored AI-produced work and are attributed accordingly. This preprint has not undergone conventional human peer review or refereeing at publication. Automated adversarial reviews recorded in the package are not human peer review. The upstream universal-optimality theorem remains the explicit external mathematical dependency of the conclusion.

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